Yiduo Qu

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EDUCATION

University of Cambridge (St Edmund's College) **Oct. 2026 – Jun. 2030**
BA/MEng (Hons), Engineering Cambridge, UK

- Awarded the SJIPF – St Edmund's College Merit Scholarship for the full duration of studies

St. Joseph's Institution International (SJII) **Jan. 2018 – Dec. 2023**
International Baccalaureate: 44/45; IGCSE: 9A* Singapore

- Top 3 in Cohort for Overall Academic Excellence; Dean's List (Top 3) for Physics and Chemistry
- Vice-President of Senior Council (Student Government), representing 1,000+ students

PUBLICATIONS

G. Chen*, **Y. Qu***, Q. Xia*, et al. DiagGen: Agentic Generation of Deformable Assets with Sim-based Diagnostics for Robotic Simulation. *CoRL* 2026 Workshop. arXiv:2609.23103 · Project page (*equal contribution)

P. Yang, X. Wang, P. Jing, K. Wen, **Y. Qu**, et al. RARM: Confidence-Gated Progress Reward Modeling for RL in Manipulation. *CoRL* 2026. arXiv:2606.22027 · Project page

EXPERIENCE

Human-Centered Robotics Lab (National University of Singapore) **Feb. 2026 – Sep. 2026**
Full-time Researcher Singapore

RARM: Learning robotic manipulation from a single demonstration

- Trained policies from scratch with RL (PPO, SAC, DrQ-v2) across 8 tasks in MetaWorld and LIBERO, totalling ~2,000 GPU-hours on GH200 nodes.
- Fine-tuned a DINOv3 vision foundation model + LoRA reward model on ~500 unlabelled videos and one demo per task; outperformed 7 baselines.
- Implemented a VR teleoperation pipeline (Meta Quest 3, ROS2) to collect robot data on a bimanual setup; ran 200+ hours of real-world experiments.

DiagGen: Turning one phone photo into a simulation-ready deformable object

- Built the agentic generation pipeline (SAM3, OmniPart, material inference, mesh processing).
- Designed diagnostic agent that probes assets with simulated grasps and routes repair to the stage at fault.
- Built a 40-asset dataset and PAIP, a VLM-judge metric scoring physical plausibility of assets.

Singapore Police Force **Jan. 2024 – Nov. 2025**
Officer Cadet (1 yr), National Service Inspector (1 yr) — mandatory national service Singapore

- Supervised 15+ auxiliary police officers, managing rotation, training and onboarding at Tanglin Division HQ.
- Planned and led security deployments around the Tanglin area for 20+ foreign VVIPs during diplomatic visits in 2025.
- Executed security deployments for General Election 2025 and Shangri-La Dialogue 2025.

Robotics Unbound (Robotics Outreach NGO) **Jan. 2021 – Jun. 2024**
Director Singapore

- Awarded the National Youth Achievement Award (Gold) for outstanding contribution to community service.
- Developed a Scratch curriculum and taught LEGO WeDo robotics to 400+ children across 10+ children's homes.
- Managed and trained a team of 30 volunteers in programming, robotics and teaching children aged 9–13.

SKILLS

- **Technical:** Python, C++, C, Bash, Linux, Docker, SLURM/HPC (multi-GPU), Git
- **ML/Robotics:** PyTorch, Stable-Baselines3, NumPy, TensorBoard; ROS2, MuJoCo, MetaWorld, LIBERO
- **Languages:** English (fluent); Mandarin (fluent)